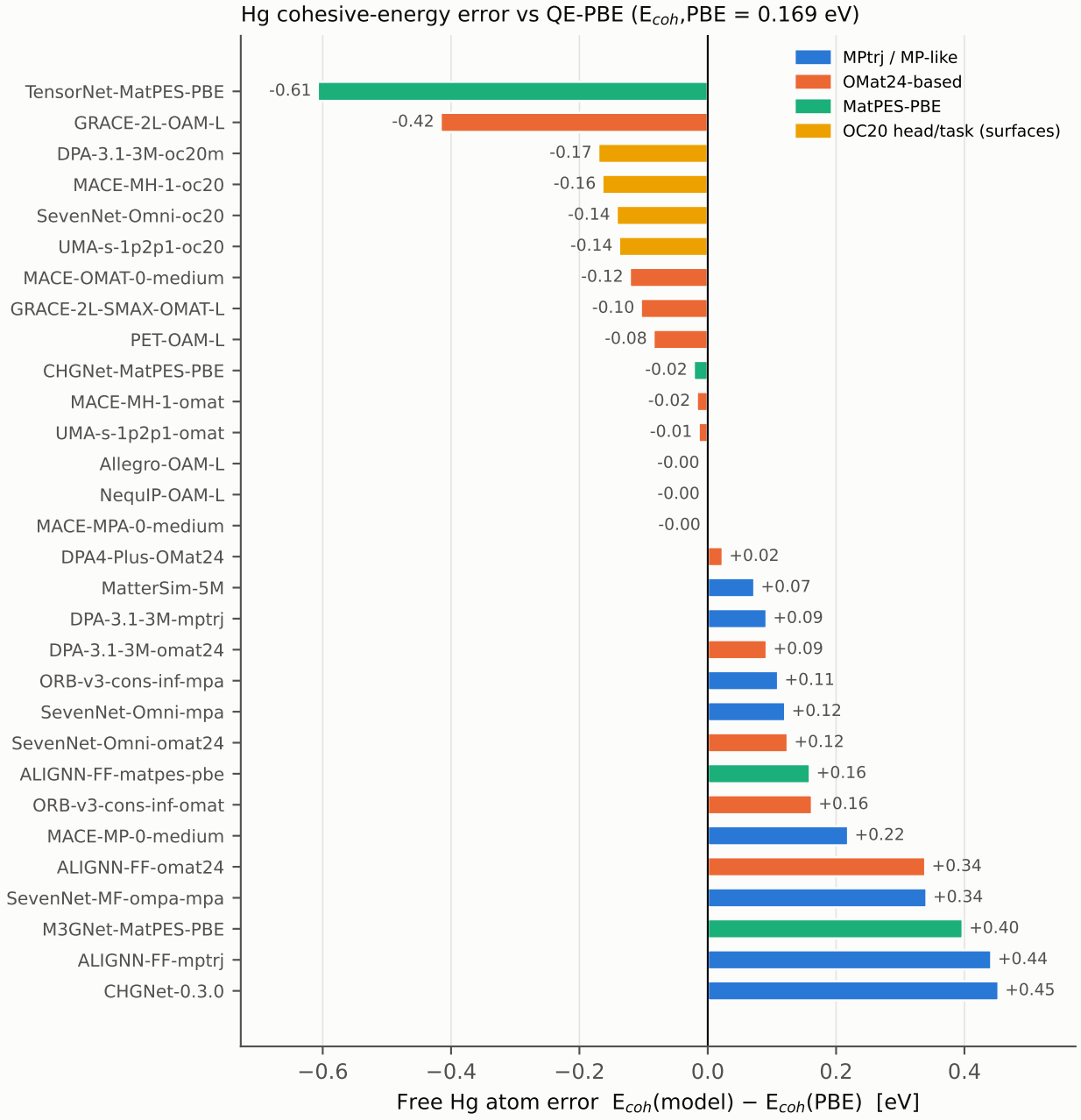
Hg cohesive-energy error vs QE-PBE (E_{coh},PBE = 0.169 eV)
MPtrj / MP-like
OMat24-based
MatPES-PBE
OC20 head/task (surfaces)
TensorNet-MatPES-PBE -0.61
GRACE-2L-OAM-L -0.42
DPA-3.1-3M-oc20m -0.17
MACE-MH-1-oc20 -0.16
SevenNet-Omni-oc20 -0.14
UMA-s-1p2p1-oc20 -0.14
MACE-OMAT-0-medium -0.12
GRACE-2L-SMAX-OMAT-L -0.10
PET-OAM-L -0.08
CHGNet-MatPES-PBE -0.02
MACE-MH-1-omat -0.02
UMA-s-1p2p1-omat -0.01
Allegro-OAM-L -0.00
NequIP-OAM-L -0.00
MACE-MPA-0-medium -0.00
DPA4-Plus-OMat24 +0.02
MatterSim-5M +0.07
DPA-3.1-3M-mptrj +0.09
DPA-3.1-3M-omat24 +0.09
ORB-v3-cons-inf-mpa +0.11
SevenNet-Omni-mpa +0.12
SevenNet-Omni-omat24 +0.12
ALIGNN-FF-matpes-pbe +0.16
ORB-v3-cons-inf-omat +0.16
MACE-MP-0-medium +0.22
ALIGNN-FF-omat24 +0.34
SevenNet-MF-ompa-mpa +0.34
M3GNet-MatPES-PBE +0.40
ALIGNN-FF-mptrj +0.44
CHGNet-0.3.0 +0.45
−0.6
−0.4
−0.2
0.0
0.2
0.4
Free Hg atom error E_{coh}(model) − E_{coh}(PBE) [eV]